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Assignment Statement:

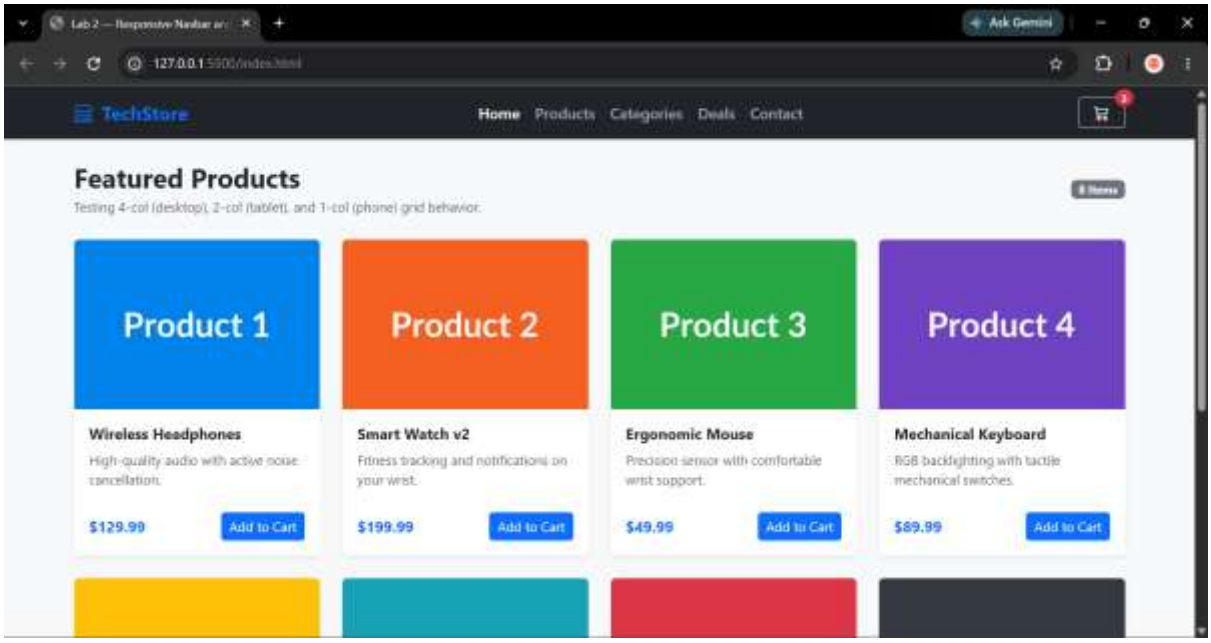
Difficulty: Easy–Medium · **Covers:** Lectures 7–10 · **Folder:** lab-2/

A focused, smaller-scale rehearsal for Assignment 2: build **just** a responsive navbar (logo left, links center, cart icon right, collapsing into a hamburger menu below 768px) and a 4-column-desktop / 2-column-tablet / 1-column-phone product grid, using Flexbox or Grid plus Bootstrap or Tailwind. Content can be placeholder text/images — the point is the responsive mechanics, not the final design.

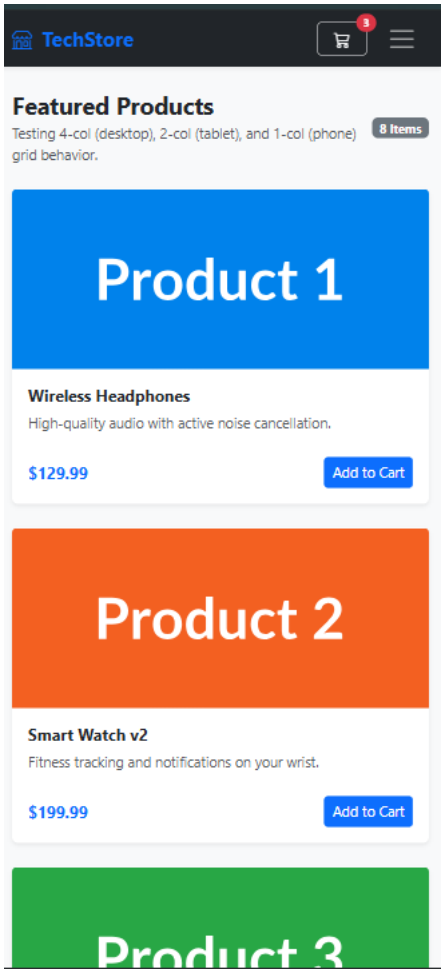
Deliverables in lab-2/: index.html, your CSS/framework files, and your completion PDF (with screenshots at phone/tablet/desktop widths).

Output:

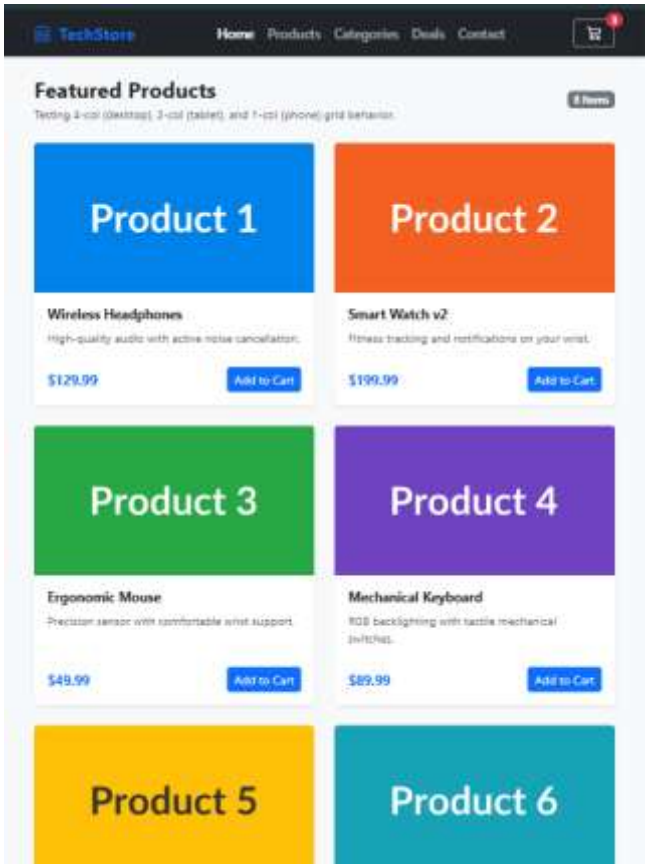
Desktop Screenshot:



Mobile Screenshot:



Tablet Screenshot:



Code:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Lab 2 – Responsive Navbar and Grid Drill</title>
  <!-- Bootstrap 5 CSS -->
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <!-- Bootstrap Icons -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-
icons@1.11.3/font/bootstrap-icons.min.css" rel="stylesheet">
  <!-- Custom CSS -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <!-- RESPONSIVE NAVBAR -->
```

```
  <nav class="navbar navbar-expand-md navbar-dark bg-dark sticky-top shadow-sm
py-2">
    <div class="container">

      <!-- Logo (Left) -->
      <a class="navbar-brand fw-bold text-primary" href="#">
        <i class="bi bi-shop me-1"></i> TechStore
      </a>

      <!-- Right Controls: Cart Icon & Hamburger Toggle -->
      <div class="d-flex align-items-center gap-2 order-md-3">
        <a href="#" class="btn btn-outline-light position-relative btn-sm px-
3">
          <i class="bi bi-cart3 fs-6"></i>
          <span class="position-absolute top-0 start-100 translate-middle
badge rounded-pill bg-danger">
            3
          </span>
        </a>

        <!-- Hamburger Button (< 768px) -->
        <button class="navbar-toggler border-0" type="button" data-bs-
toggle="collapse" data-bs-target="#navMenu" aria-controls="navMenu" aria-
expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
      </div>

      <!-- Collapsible Nav Links (Center) -->
      <div class="collapse navbar-collapse justify-content-center order-md-2
my-2 my-md-0" id="navMenu">
        <ul class="navbar-nav mb-2 mb-md-0 fw-semibold text-center">
          <li class="nav-item"><a class="nav-link active"
href="#">Home</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Products</a></li>
          <li class="nav-item"><a class="nav-link"
href="#">Categories</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Deals</a></li>
          <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
        </ul>
      </div>

    </div>
  </nav>

  <!-- MAIN CONTENT: RESPONSIVE GRID -->
  <main class="container py-4">
    <div class="d-flex justify-content-between align-items-center mb-4">
```

```
      <div>
        <h1 class="h3 fw-bold mb-1">Featured Products</h1>
        <p class="text-muted small mb-0">Testing 4-col (desktop), 2-col
(tablet), and 1-col (phone) grid behavior.</p>
      </div>
      <span class="badge bg-secondary">8 Items</span>
    </div>

    <!-- Grid Mechanics:
         row-cols-1  -> 1 Column on Phone (< 768px)
         row-cols-md-2 -> 2 Columns on Tablet (>= 768px)
         row-cols-lg-4 -> 4 Columns on Desktop (>= 992px)
    -->
    <div class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-4">

      <!-- Product Card 1 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/0082eb/ffffff?text=Product+1"
class="card-img-top" alt="Product 1">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">Wireless Headphones</h5>
            <p class="card-text text-muted small flex-grow-1">High-quality
audio with active noise cancellation.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$129.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

      <!-- Product Card 2 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/f36021/ffffff?text=Product+2"
class="card-img-top" alt="Product 2">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">Smart Watch v2</h5>
            <p class="card-text text-muted small flex-grow-1">Fitness tracking
and notifications on your wrist.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$199.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
```

```
        </div>
      </div>

      <!-- Product Card 3 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/28a745/ffffff?text=Product+3"
class="card-img-top" alt="Product 3">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">Ergonomic Mouse</h5>
            <p class="card-text text-muted small flex-grow-1">Precision sensor
with comfortable wrist support.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$49.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

      <!-- Product Card 4 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/6f42c1/ffffff?text=Product+4"
class="card-img-top" alt="Product 4">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">Mechanical Keyboard</h5>
            <p class="card-text text-muted small flex-grow-1">RGB backlighting
with tactile mechanical switches.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$89.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

      <!-- Product Card 5 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/ffc107/333333?text=Product+5"
class="card-img-top" alt="Product 5">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">Portable Speaker</h5>
            <p class="card-text text-muted small flex-grow-1">Waterproof
Bluetooth speaker with deep bass.</p>
```

```
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$59.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

      <!-- Product Card 6 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/17a2b8/ffffff?text=Product+6"
class="card-img-top" alt="Product 6">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">HD Webcam 1080p</h5>
            <p class="card-text text-muted small flex-grow-1">Clear video
calls with built-in microphone.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$39.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

      <!-- Product Card 7 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
          <img src="https://placehold.co/400x300/dc3545/ffffff?text=Product+7"
class="card-img-top" alt="Product 7">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">USB-C Power Bank</h5>
            <p class="card-text text-muted small flex-grow-1">20,000mAh
capacity with fast charging support.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$34.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

      <!-- Product Card 8 -->
      <div class="col">
        <div class="card h-100 border-0 shadow-sm product-card">
```

```
          <img src="https://placehold.co/400x300/343a40/ffffff?text=Product+8"
class="card-img-top" alt="Product 8">
          <div class="card-body d-flex flex-column">
            <h5 class="card-title h6 fw-bold">Monitor Stand Riser</h5>
            <p class="card-text text-muted small flex-grow-1">Ergonomic desk
organizer with storage slots.</p>
            <div class="d-flex justify-content-between align-items-center mt-
2">
              <span class="fw-bold text-primary">$29.99</span>
              <button class="btn btn-sm btn-primary">Add to Cart</button>
            </div>
          </div>
        </div>
      </div>

    </div>
  </main>

  <!-- FOOTER -->
  <footer class="bg-dark text-light py-3 mt-5 border-top">
    <div class="container text-center small text-muted">
      <p class="mb-0">&copy; 2026 Lab 2 Drill — Responsive Navbar & Grid
mechanics.</p>
    </div>
  </footer>

  <!-- Bootstrap 5 JS Bundle -->
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min
.js"></script>
</body>
</html>
```

styles.css:

```
body {
  background-color: #f8f9fa;
  font-family: system-ui, -apple-system, sans-serif;
}

.product-card {
  transition: transform 0.2s ease-in-out, box-shadow 0.2s ease-in-out;
}

.product-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 8px 16px rgba(0, 0, 0, 0.1) !important;
}

.product-card img {
```

```
  object-fit: cover;
  height: 180px;
}
```